

## Unit 108 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 63.3 ac, BAF 20 variable-radius cruise on a 300 ft grid, 31 plots, 344 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

**Table 1. Stand summary from the BAF 20 cruise, per acre**

| STAND DENSITY        |                                    | STAND INFO          |                                                |
|----------------------|------------------------------------|---------------------|------------------------------------------------|
| Basal area           | 222 sq ft/ac (SE 2.6 %)            | Type                | Sierran mixed conifer                          |
| Trees per acre       | 68                                 | CWHR size / density | 5 / D (cover 91 %)                             |
| QMD                  | 24.5 in                            | SDI                 | 274 (47 % of max 580)                          |
| SAWTIMBER (>= 10 in) |                                    | BIOMASS (< 10 in)   |                                                |
| BA   QMD             | 222 sq ft/ac   24.5 in             | BA   QMD            | 0 sq ft/ac   0.0 in                            |
| TPA                  | 68                                 | TPA                 | 0                                              |
| Volume               | 39.2 MBF/ac (7810 cu ft)           | Volume              | 0.0 green tons/ac (0 cu ft)                    |
| TREATMENT TARGET     |                                    | CRUISE DESIGN       |                                                |
| Leave                | 35 % of max SDI = ~164 sq ft/ac BA | Sampling error      | 5.4 % at 95 % (t = 2.04, n = 31)               |
| Species (BA)         | WF 68, PP 66, IC 36, DF 29, SP 22  | Stratum standard    | 40 % - MEETS; 31 plots vs 20-plot practice: ok |

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1). Sources and URLs: cruise\_data.xlsx, Standards and assumptions.

**Table 2. Stand table by 4-inch DBH class, per acre**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 8-11.9        | 4          | 3.6         | 2.6          | 66          | 0.3         |
| 12-15.9       | 22         | 13.2        | 14.2         | 345         | 1.7         |
| 16-19.9       | 33         | 12.0        | 21.3         | 610         | 3.1         |
| 20-23.9       | 53         | 12.7        | 34.2         | 1063        | 5.3         |
| 24-27.9       | 60         | 10.7        | 38.7         | 1307        | 6.6         |
| 28-31.9       | 44         | 5.9         | 28.4         | 1021        | 5.1         |
| 32-35.9       | 39         | 4.0         | 25.2         | 919         | 4.6         |
| 36-39.9       | 35         | 2.9         | 22.6         | 896         | 4.5         |
| 40-43.9       | 26         | 1.7         | 16.8         | 701         | 3.5         |
| 44-47.9       | 6          | 0.3         | 3.9          | 167         | 0.8         |
| 48-51.9       | 7          | 0.3         | 4.5          | 212         | 1.1         |
| 52-55.9       | 6          | 0.2         | 3.9          | 209         | 1.1         |
| 56-59.9       | 5          | 0.2         | 3.2          | 153         | 0.8         |
| 60-63.9       | 4          | 0.1         | 2.6          | 140         | 0.7         |
| <b>All</b>    | <b>344</b> | <b>68.1</b> | <b>221.9</b> | <b>7810</b> | <b>39.2</b> |

**Table 3. Stock table by species, per acre**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| WF         | 106        | 22.5        | 68.4         | 23.6        | 2344        | 11.8        |
| PP         | 103        | 19.0        | 66.5         | 25.3        | 2641        | 13.3        |
| IC         | 56         | 12.9        | 36.1         | 22.7        | 993         | 5.0         |
| DF         | 45         | 7.0         | 29.0         | 27.6        | 1002        | 5.0         |
| SP         | 34         | 6.7         | 21.9         | 24.4        | 830         | 4.2         |
| <b>All</b> | <b>344</b> | <b>68.1</b> | <b>221.9</b> | <b>24.5</b> | <b>7810</b> | <b>39.2</b> |

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 108: plots on the 300 ft grid (orange = QA flag)

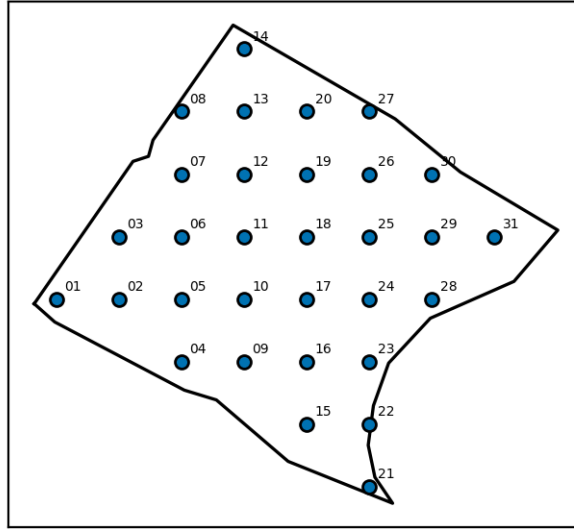


Figure 1. Plot locations with QA status.

**QA findings:** no issues found; unit accepted

Table 4. Plot list

| Plot   | Trees in | BA sq ft/ac | Slope % | Status |
|--------|----------|-------------|---------|--------|
| 108-01 | 10       | 200         | 11      | ok     |
| 108-02 | 11       | 220         | 22      | ok     |
| 108-03 | 9        | 180         | 23      | ok     |
| 108-04 | 10       | 200         | 18      | ok     |
| 108-05 | 10       | 200         | 16      | ok     |
| 108-06 | 12       | 240         | 21      | ok     |
| 108-07 | 11       | 220         | 27      | ok     |
| 108-08 | 11       | 220         | 23      | ok     |
| 108-09 | 11       | 220         | 27      | ok     |
| 108-10 | 12       | 240         | 12      | ok     |
| 108-11 | 11       | 220         | 14      | ok     |
| 108-12 | 14       | 280         | 25      | ok     |
| 108-13 | 6        | 120         | 17      | ok     |
| 108-14 | 13       | 260         | 6       | ok     |
| 108-15 | 14       | 280         | 15      | ok     |
| 108-16 | 10       | 200         | 13      | ok     |
| 108-17 | 11       | 220         | 9       | ok     |
| 108-18 | 11       | 220         | 8       | ok     |
| 108-19 | 9        | 180         | 32      | ok     |
| 108-20 | 12       | 240         | 17      | ok     |
| 108-21 | 10       | 200         | 22      | ok     |
| 108-22 | 12       | 240         | 30      | ok     |
| 108-23 | 11       | 220         | 18      | ok     |
| 108-24 | 12       | 240         | 11      | ok     |
| 108-25 | 10       | 200         | 16      | ok     |
| 108-26 | 13       | 260         | 15      | ok     |
| 108-27 | 13       | 260         | 19      | ok     |
| 108-28 | 11       | 220         | 10      | ok     |
| 108-29 | 11       | 220         | 13      | ok     |
| 108-30 | 10       | 200         | 20      | ok     |
| 108-31 | 13       | 260         | 10      | ok     |